

HWA CHONG INSTITUTION



C2 PRELIMINARY EXAMINATION

H1 CHEMISTRY 8872/01

PAPER 1

Multiple Choice

23 September 2011

50 mins

Do not open this booklet until you are told to do so.

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Complete the information on the optical mark sheet (OMS) as shown below.

1. Enter your NAME (as in NRIC). _____

Write your name

2. Enter the SUBJECT TITLE. _____

3. Enter the PAPER NUMBER. _____

4. Enter your CT GROUP. _____

Write your CT group

5. Date. _____

6. Enter your NRIC NUMBER or FIN NUMBER.

7. Now SHADE the corresponding lozenge in the grid for EACH DIGIT or LETTER

NRIC / FIN															
S	0	0	0	0	0	0	0	0	0	A	K	U			
F	1	1	1	1	1	1	1	1	1	B	L	V			
G	2	2	2	2	2	2	2	2	2	C	M	W			
T	3	3	3	3	3	3	3	3	3	D	N	X			
	4	4	4	4	4	4	4	4	4	E	O	Y			
	5	5	5	5	5	5	5	5	5	F	P	Z			
	6	6	6	6	6	6	6	6	6	G	Q				
	7	7	7	7	7	7	7	7	7	H	R				
	8	8	8	8	8	8	8	8	8	I	S				
	9	9	9	9	9	9	9	9	9	J	T				

Write and
shade your
NRIC
or FIN number

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

Each correct answer will score one mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 How many oxygen atoms are there in 10 g of ascorbic acid, $C_6H_8O_6$?

A 3.4×10^{22}
C 3.4×10^{23}

B 2.0×10^{23}
D 2.0×10^{24}

- 2 10 cm^3 of a hydrocarbon, C_xH_y , was exploded with an excess of oxygen. There was a contraction of 35 cm^3 . When the product was treated with aqueous sodium hydroxide, there was a further contraction of 40 cm^3 .

What is the molecular formula of the hydrocarbon? [All gas volumes were measured at r.t.p.]

A C_3H_6

B C_3H_8

C C_4H_8

D C_4H_{10}

- 3 In which of the following reactions is the acid acting as an oxidant?

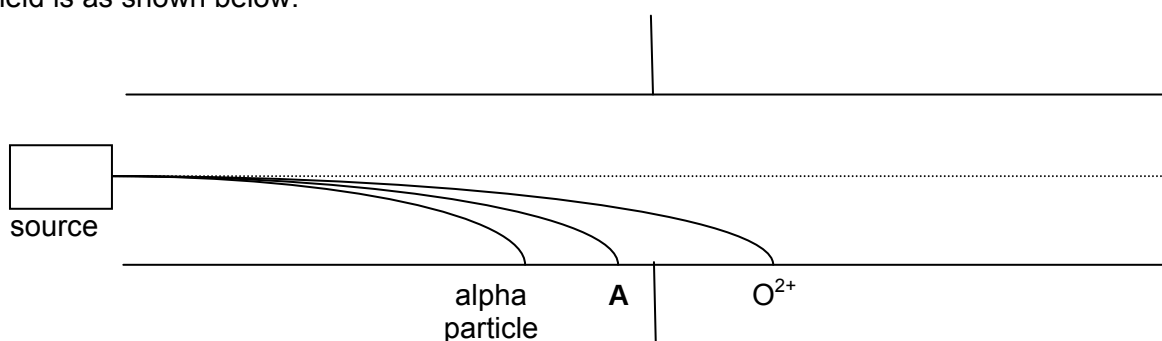
A $KBr + H_3PO_4 \rightarrow HBr + KH_2PO_4$

B $MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$

C $12HClO_4 + P_4O_{10} \rightarrow 6Cl_2O_7 + 4H_3PO_4$

D $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$

- 4 Alpha particles, He^{2+} , are commonly emitted by larger radioactive nuclei. The path of a mixture of an unknown ion, **A**, O^{2+} and alpha particles after passing through an electric field is as shown below.



Which of the following represents **A**?

A Be^{2+}

B C^+

C Al^{3+}

D Cl^+

5 Two elements **B** and **C** have the following properties.

- **B** and **C** react with sodium to form ionic compounds Na_2B and Na_2C respectively.
- When reacted with fluorine, **B** forms only BF_2 , while **C** forms CF_2 and CF_6 .

Which of the following pairs of electronic configurations for **B** and **C** is correct?

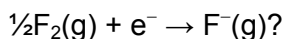
	B	C
A	$1s^2 2s^2 2p^4$	$1s^2 2s^2 2p^2$
B	$1s^2 2s^2 2p^2$	$1s^2 2s^2 2p^6 3s^2 3p^2$
C	$1s^2 2s^2 2p^2$	$1s^2 2s^2 2p^6 3s^2 3p^3$
D	$1s^2 2s^2 2p^4$	$1s^2 2s^2 2p^6 3s^2 3p^4$

6 The Voyager 2 probe has shown that the surface of Triton, a moon of the planet Neptune, contains solid methane which flows rapidly.

Which statement best explains the mobility of solid methane?

- A** Triton has a low atmospheric pressure.
- B** Solid methane has a giant covalent structure.
- C** Methane molecules contain strong C-H bonds.
- D** Methane molecules have weak intermolecular forces.

7 Which of the following gives the magnitude of the enthalpy change for the reaction:

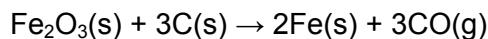


- A** the atomisation energy of fluorine
- B** the first electron affinity of fluorine
- C** the sum of the bond energy and first electron affinity of fluorine
- D** the sum of the atomisation energy and first electron affinity of fluorine

8

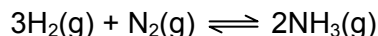
Compound	$\Delta H_f / \text{kJ mol}^{-1}$
$\text{Fe}_2\text{O}_3(\text{s})$	-825
$\text{CO}(\text{g})$	-110

Given the above information, what is the enthalpy change (in kJ mol^{-1}) for the following reaction?



- A** -715
- B** +495
- C** +715
- D** +1155

- 9 The following shows the equilibrium reaction between hydrogen and nitrogen in the Haber process.

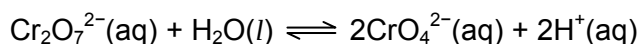


The forward and backward rate constants are given as k_1 and k_{-1} respectively.

What will happen to k_1 , k_{-1} , and the equilibrium constant, K_p , if the volume of the reaction vessel is decreased at constant temperature?

	k_1	k_{-1}	K_p
A	unchanged	unchanged	unchanged
B	unchanged	increases	unchanged
C	increases	increases	increases
D	increases	increases	decreases

- 10 Which statement concerning the following equilibrium reaction is true?



- A It involves a redox reaction.
- B The equilibrium constant, K_c , has no units.
- C A decrease in pH will result in an increase in $[\text{Cr}_2\text{O}_7^{2-}(\text{aq})]$.
- D Addition of a catalyst will result in an increase in $[\text{Cr}_2\text{O}_7^{2-}(\text{aq})]$.

- 11 Which of the following pairs of solutions would form an acidic buffer when mixed together?

- | | | | |
|---|---------------|---|--|
| A | HCN and NaCN | B | HNO ₃ and NaNO ₃ |
| C | NaOH and NaCl | D | HCl and NaOH |

- 12 The rate of reaction of a strip of magnesium ribbon and 50 cm³ of 1.0 mol dm⁻³ H₂SO₄ is determined at 25 °C.

In which of the following cases would *both* the new conditions contribute to the increase in the rate of reaction?

- A Mg powder and 100 cm³ of 1.0 mol dm⁻³ H₂SO₄
- B Mg powder and 50 cm³ of 0.8 mol dm⁻³ H₂SO₄
- C 25 cm³ of 2.0 mol dm⁻³ H₂SO₄ at 30 °C
- D 100 cm³ of 1.0 mol dm⁻³ H₂SO₄ at 30 °C

- 13 The rate of removal of the pain-killing drug paracetamol from the body is a first order reaction with a rate constant, k , of 0.26 h^{-1} .

How long will it take for 50% of the paracetamol ingested to remain in the body?

- A 1.3 h B 2.7 h C 4.0 h D 5.4 h

- 14 The properties of the oxides of three Period 3 elements, E, F and G, are given below.

Oxide of E is amphoteric.

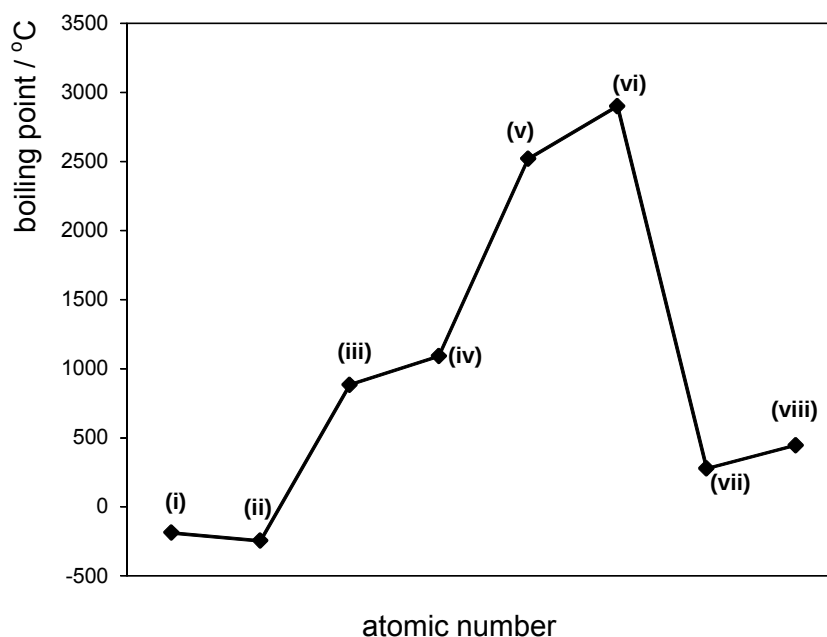
Oxide of F reacts with aqueous sodium hydroxide at room temperature.

Oxide of G dissolves in water to form a strong alkaline solution.

Which of the following is correct in order of increasing proton number of the elements?

- A G, E, F B E, F, G C G, F, E D F, G, E

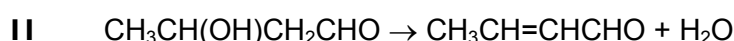
- 15 The following graph shows the *boiling points* of eight consecutive elements, (i) to (viii), in Periods 2 and 3 of the Periodic Table.



Which of the following statements about the oxides of the elements is **false**?

- A (i) forms an oxide which melts at 1286°C .
 B (v) forms an oxide which can react with bases.
 C (vi) forms a solid oxide.
 D (viii) forms more than one oxide.

- I $2\text{CH}_3\text{CHO} \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CHO}$
aldol



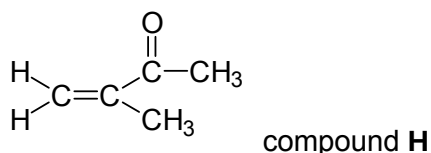
Which of the following correctly describes reactions I and II?

- | | I | II |
|---|--------------|-------------|
| A | elimination | reduction |
| B | substitution | elimination |
| C | addition | reduction |
| D | addition | elimination |

- 18** What is the total number of structural and cis-trans isomers for a compound with molecular formula $\text{C}_3\text{H}_5\text{F}$, excluding cyclic structures?

- A** 3 **B** 4 **C** 5 **D** 6

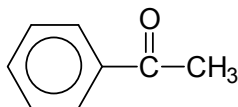
- 19** Compound **H** can be used to make the hard outer covering of golf balls.



Which of the following is true about **H**?

- A** It can exist as cis-trans isomers.
- B** It can react with Fehling's solution.
- C** One mole of **H** can react exactly with 2 moles of hydrogen gas.
- D** It can react with ethanol in the presence of a little concentrated H_2SO_4 on heating.

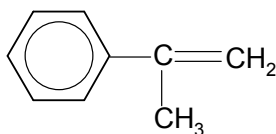
- 20 Which of the reactions does the following molecule **not** undergo?



- A** addition
B hydrolysis
C reduction
D substitution
- 21 Which of the following is the major product formed when 1-chloropropane is heated with ethanolic sodium hydroxide?

- A** $\text{CH}_3\text{CH}=\text{CH}_2$
B $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
C $\text{CH}_3\text{CH}_2\text{CH}_2\text{ONa}$
D $\text{CH}_3\text{CH}_2\text{CH}_2\text{OCH}_2\text{CH}_3$

- 22 Which of the following represents the final product obtained when the alkene below reacts with cold, alkaline manganate(VII) followed by heating with acidified dichromate(VI)?



- A** CC(=O)c1ccccc1
B OC(=O)c1ccccc1
C CC(=O)C(O)c1ccccc1
D OC(=O)C(O)c1ccccc1

- 23 Which observation is true about the reactions of $\text{CH}_3\text{CH}(\text{CH}_3)\text{CHO}$?

- A** an effervescence with sodium metal
B fumes of hydrogen chloride with PCl_5
C an orange precipitate on warming with 2,4-dinitrophenylhydrazine
D a pale yellow precipitate on warming with alkaline aqueous iodine

- 24 Four drops of 1-chlorobutane, 1-bromobutane and 1-iodobutane were separately added to three test-tubes containing 1.0 cm^3 of aqueous silver nitrate at 60°C . The following reaction occurred.



Which of the following best explains why the rate of formation of cloudiness (precipitate) in the tubes was in the order $\text{RCI} < \text{RBr} < \text{RI}$?

- A The R-X bond polarity decreases from RCl to RI.
 B The bond energy of R-X decreases from RCl to RI.
 C The solubility of AgX(s) decreases from AgCl to AgI.
 D The ionisation energy of the halogen decreases from Cl to I.
- 25 Compound J is subjected to two separate tests and the results were as shown below.

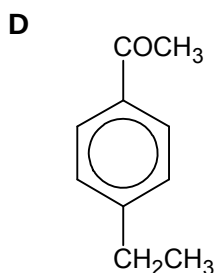
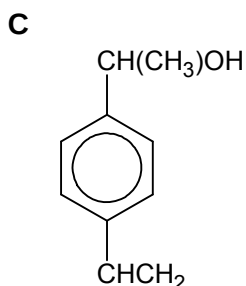
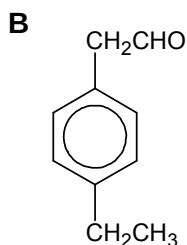
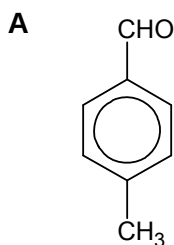
Reagents and conditions
 Test 1: Acidified KMnO_4 and heat

Observations
 Purple solution turns colourless
 Effervescence of colourless gas that turns limewater chalky

Test 2: Tollen's reagent and warm

Silver mirror formed

Which of the following could be compound J?



Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

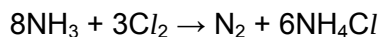
Decide whether each of the statements is or is not correct (you may find it helpful to place a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** The following shows ammonia and chlorine reacting in the gaseous phase.



Which statements are correct?

- 1** Ammonia behaves as a reducing agent.
- 2** Ammonia behaves as a base.
- 3** The oxidation number of the hydrogen changes.

- 27** Which of the following are true about metallic bonding?

- 1** The larger the size of the metal cation, the stronger is the metallic bonding.
- 2** The greater the charge of the metal cation, the stronger is the metallic bonding.
- 3** The greater the number of valence electrons the metal has, the stronger is the metallic bonding.

- 28** 0.1 mole of each of the following is separately added to 100 cm³ of water.

Which of the following show the resulting solutions in an increasing order of pH values?

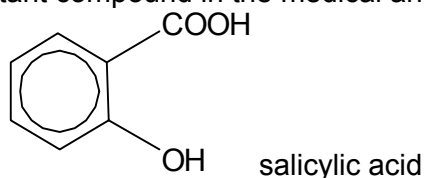
- 1** PCl_3 , AlCl_3 , NaCl
- 2** NH_3 , NaOH , $\text{Ba}(\text{OH})_2$
- 3** HCl , $\text{CH}_3\text{CO}_2\text{H}$, $\text{CH}_3\text{CH}_2\text{OH}$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

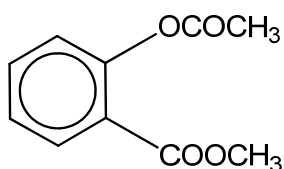
No other combination of statements is used as a correct response.

- 29** Salicylic acid is an important compound in the medical and cosmetic industries.

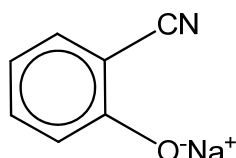


Which compounds would give salicylic acid on acid hydrolysis?

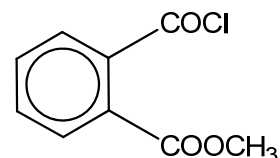
1



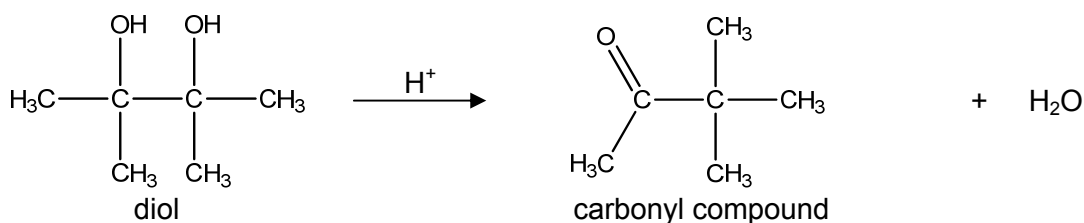
2



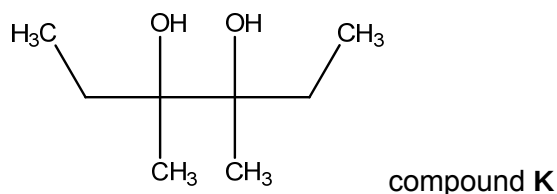
3



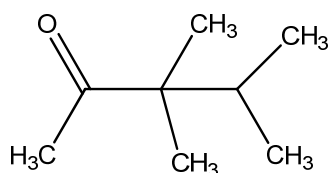
- 30** The pinacol rearrangement involves the reaction of a diol in acidic conditions to form a carbonyl compound and is shown below.



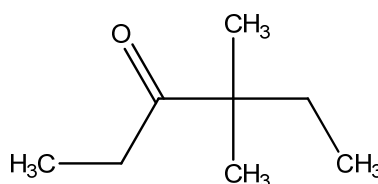
Which of the following are the products of the pinacol rearrangement of compound **K**?



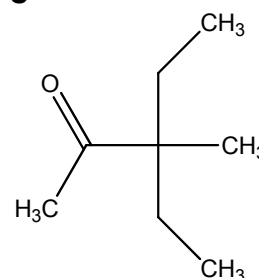
1



2



3



~End of Paper 1~